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Welcome to the Chameleon Project

Chameleon is a versatile emulator for contactless smartcards. It is meant to be a programmable platform to assess security aspects in RFID environments and can be used in different attack scenarios like replay or relay attacks and even for emulating an existing smartcard, thus behaving as a perfect clone of a given card.

The Chameleon Project has been started by the [Chair for Embedded Security](#) at the German University in Bochum and has been licensed as open source to let everyone benefit from the work that has been done so far. We are always looking forward to any means of contributing or contacting us.

For a small demonstration of the Chameleon's capabilities, watch the 29C3 talk "Milking the Digital Cash Cow" by Timo Kasper linked below.

Chameleon-Mini

The newest version of the Chameleon is the Chameleon-Mini. It is a small credit-card shaped and feature-packed device cheaply available for everyone.



The whole project is published under a open-source license. See the code repository for all the files that are freely available.

Features:

- Firmware support for ISO14443A Codec
- Firmware support for Mifare Classic 1K and 4K emulation
- Hardware support for ASK modulation (Both 10% and 100%) to cover almost any card standard available
- Hardware support for ASK and BPSK load modulation using a subcarrier
- Modular firmware structure allows for easy expandability of other cards and standards
- Support for quick and reliable firmware update via Atmel DFU bootloader, thus programming hardware is required only once
- Can be controlled using a fully documented AT-like command set via CDC
- 1MByte of card memory allows for multiple card emulations to reside on the device simultaneously

- These settings can be switched even without a USB connection, just by pressing the button
- Card contents can be easily uploaded and downloaded by means of the command line and X-MODEM
- Thus allowing the Chameleon to be interfaced with standard terminal software as well as user written scripts and applications

Related Publications:

- [Paper describing the "Chameleon"](#)
- [Security Analysis Of Pervasive Wireless Devices](#)
- [29c3: Milking the Digital Cash Cow](#)
- [An Embedded System for Practical Security Analysis of Contactless Smartcards](#)

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